ME 685 Mobile Microrobotic Systems

Spring Term 2025

Department of Mechanical Engineering
Stevens Institute of Technology

PROJECT MENTOR: Yong Shi (yongshi@mit.edu)

Earthworm inspired Soft Body Robot

Problem Statement

Robots using peristaltic locomotion can navigate through narrow space in a complex environment and have great potential for aerospace and medical applications, such as cleaning up narrow channels, repairing mechanical and electric interfaces or connections. This project involves the design, fabrication and testing of such a robot which may have segmented tubular structures being able to perform bio-inspired peristalsis for locomotion. Locomotion of the robots might be achieved by shape memory alloy, shape memory polymer or piezoelectric actuators.

You can choose to focus on four of the following aspects:

1) Scaling law for dominate micro forces and design strategy to overcome
2) Locomotion mechanisms and actuator design
3) Contact force and obstacle/distance sensing
4) Power requirement and supply
5) Microrobot control and localization
6) Fabrication process
7) Testing and scaling-up model

Design Specifications

The specific goal of this project is to design a biomimetic locomotion mechanism for micro scale robot which can navigate through tiny space at low Reynolds number. The overall device diameter should be smaller than 1 mm. The operation voltage should be smaller than 10 V. Wireless power supply and communication should be evaluated.

Fabrication process should be designed, and the process should be compatible with the system specifications. The fabrication process should be assumed to be well-acceptable with reasonable manufacturing tolerances.

Overview

The team should work together to develop a research/design block diagram of tasks to reach the design goal. All these tasks interact, so the goal of the team is to delegate tasks and provide communication between tasks so that the final merged device design meets the requirements.

First-Order Device Concept

Survey the literature and explore micro/nano earthworm like robots. Create simple but appropriate models to evaluate/simulate the locomotion principle, and make sure the designs are compatible with micro fabrication. The team can also use an equivalent-circuit model to predict system level behavior or performance of the robot. Powering, communication and control principles should also be discussed. It's required for the team to come up with two or three different design concepts and choose the most promising one.

Fabrication and Packaging

Develop a fabrication process flow for the proposed design.

Modeling

Use analytical tools/equations introduced in the class, finite element analysis of individual modules such as locomotion, powering or a combination of lumped elements models or Simulink for micro force analysis, micro robot design and control optimization. Evaluate the effects of any error in fabrication, control, or power supply would have on the behavior of the system.

Iterate the Design

It may require iteration of various details until the design fits together as a whole.